Proofs for the dog game with inertia

This note proves the two propositions of Section 5.20 of the research report. The proofs are mine and have not been refereed; every step is also checked numerically in `tests/test_inertia_theory.py` and the studies `inertia_explicit` and `inertia_random`.

Setting

Rewards are sums over the two coordinates, the dog moves coordinate by coordinate, and the action set is the product of two intervals. So a profile is a feedback equilibrium of the two-dimensional game exactly when it is one in each coordinate: a two-dimensional deviation is a pair of one-coordinate deviations whose payoffs add. Work in one coordinate from now on.

The state is the dog's position x in [0, 1]. Red and blue choose actions a_red and a_blue in [0, 1], and the next position is

f(x, a_red, a_blue) = w0 x + (1 - w0) (w a_red + (1 - w) a_blue),

with inertia w0 in [0, 1), red's weight w in (0, 1) and blue's weight 1 - w. Player i receives -(f - h_i)^2, the squared distance of the next position from its house h_i, discounted by g in [0, 1). A stationary Markov profile pi maps positions to actions, and V_i(x) is player i's discounted payoff from x under pi. The profile is a feedback (Markov-perfect) equilibrium if, at every x, each player's action maximizes -(f - h_i)^2 + g V_i(f) given the other's action. By the one-shot deviation principle (Fudenberg and Tirole, 1991), which applies because payoffs are bounded and discounted, this is the same as having no profitable one-round deviation at any state.

Let (a_red*, a_blue*) be the one-shot Nash actions and d* = w a_red* + (1 - w) a_blue* the one-shot dog. Define

k = (1 - g w0^2) / (1 - g w0), T_i = d* + k (h_i - d*), z_h(x) = w0 x + (1 - w0) d*.

k is at least 1 because w0^2 <= w0. T_i is player i's target and z_h(x) is the next position when both hold.

Lemma 1 (the value of holding)

If both players always play the one-shot actions, then z_h(x) - d* = w0 (x - d*), so the position after t + 1 rounds is d* + w0^(t+1) (x - d*), and summing the geometric series gives, with u = x - d* and e = d* - h_i,

V_i(x) = -[e^2 / (1 - g) + 2 e w0 u / (1 - g w0) + w0^2 u^2 / (1 - g w0^2)].

Lemma 2 (the one-round objective under holding)

If player i plays something else for one round and both hold afterwards, its payoff as a function of the next position z is G_i(z) = -(z - h_i)^2 + g V_i(z). Its second derivative is -2 - 2 g w0^2 / (1 - g w0^2) < 0, so it is strictly concave, and

G_i'(z) = -2 [(z - d*) / (1 - g w0^2) + (d* - h_i) / (1 - g w0)],

which is zero exactly at z = T_i. So G_i increases up to T_i and decreases after it. With the other player at its one-shot action, player i reaches the next positions in an interval of length (1 - w0) w_i (w_red = w, w_blue = 1 - w), which is increasing in its action, and holding gives z_h(x).

Proposition 1

Holding the one-shot actions at every position is a feedback equilibrium if and only if each player at the top of the box ($a_i^* = 1$) has $k(h_i - d^*) \geq w_0(1 - d^*)$, each player at the bottom ($a_i^* = 0$) has $k(d^* - h_i) \geq w_0 d^*$, and, when $w_0 > 0$, no one-shot action is strictly inside the box.

Proof. By Lemma 2 and the one-shot deviation principle, holding is optimal for player i at x exactly when $z_h(x)$ maximizes the strictly concave G_i over i's reachable interval. For a player at the top, $z_h(x)$ is the right end of that interval, so it is optimal iff $G_i'(z_h(x)) \geq 0$, that is $z_h(x) \leq T_i$, that is

$$w_0(x - d^*) \leq k(h_i - d^*).$$

The left side is largest at $x = 1$, which gives the stated condition (for $w_0 = 0$ it always holds, since a one-shot best response at the top has $h_i \geq d^*$). A player at the bottom is the mirror image, with the worst case at $x = 0$. For a player strictly inside, $z_h(x)$ is inside its interval, so holding is optimal iff $z_h(x) = T_i$. A one-shot best response strictly inside the box puts the dog on the player's house, $h_i = d^*$, so $T_i = d^*$, and the condition becomes $w_0(x - d^*) = 0$, which fails at every x other than d^* when $w_0 > 0$. Optimality at every x for every player is the stated set of conditions. ■

Dividing by $1 - d^*$ (or by d^*), the conditions say that the player's slack $s = (h_i - d^*)/(1 - d^*)$ (or $(d^* - h_i)/d^*$) is at least $\rho(w_0, g) = w_0(1 - g w_0)/(1 - g w_0^2)$. ρ rises from 0 to 1 as w_0 goes from 0 to 1 (its derivative in w_0 is $(1 - g w_0(2 - w_0))/(1 - g w_0^2)^2$, at least $(1 - g)/(1 - g w_0^2)^2 > 0$), and its derivative in g is $w_0(w_0^2 - w_0)/(1 - g w_0^2)^2 \leq 0$, so more patient players tolerate more inertia.

Proposition 2

Suppose the coordinate is straddling: after relabeling, $h_{blue} \leq w \leq h_{red}$, so the one-shot actions are $a_{red}^* = 1$, $a_{blue}^* = 0$ and $d^* = w$. Then the policy

$$a_{red}(x) = \text{clip}(1 + (T_{red} - z_h(x))/((1 - w_0) w), 0, 1), \quad a_{blue}(x) = \text{clip}(0 + (T_{blue} - z_h(x))/((1 - w_0)(1 - w)), 0, 1)$$

is a feedback equilibrium for every w_0 in $[0, 1)$ and g in $[0, 1)$.

In words, each player aims the next position at its target, as far as its weight allows, as if the other held its one-shot action.

Proof. If $w_0 = 0$ then $z_h = d^*$, both clips return the one-shot actions, and the game is the one-shot game in every round, so assume $w_0 > 0$. Note $T_{blue} \leq d^* \leq T_{red}$, because $k \geq 1$ and $h_{blue} \leq d^* \leq h_{red}$.

Regions. Let $X_{red} = d^* + (T_{red} - d^*)/w_0$ and $X_{blue} = d^* - (d^* - T_{blue})/w_0$, so that $z_h(X_{red}) = T_{red}$ and $z_h(X_{blue}) = T_{blue}$. Red eases off its edge exactly on $R = \{x : z_h(x) > T_{red}\} = (X_{red}, 1]$, blue exactly on $B = [0, X_{blue})$, and both hold on $H = [X_{blue}, X_{red}]$ intersected with $[0, 1]$. R and B do not meet, because $X_{blue} \leq d^* \leq X_{red}$, so at most one player leaves its edge at any x. The next position under the policy is

$$N(x) = z_h(x) \text{ on } H, \quad \max(T_{red}, z_h(x) - (1 - w_0) w) \text{ on } R, \quad \min(T_{blue}, z_h(x) + (1 - w_0)(1 - w)) \text{ on } B.$$

N is nondecreasing; on R, $T_{red} \leq N(x) < x$ (since $x > X_{red} \geq T_{red}$ and $z_h(x) < x$ when $x > d^*$); on B, $x < N(x) \leq T_{blue}$. H is invariant: for x in H, $N(x) = z_h(x)$ lies between x and d^*, both in H. The targets lie in H when they lie in $[0, 1]$: $T_{red} \leq X_{red}$ because $T_{red} - d^* \geq 0$ and $w_0 \leq 1$, and $T_{red} \geq d^* \geq X_{blue}$; similarly for T_{blue}.

Values. Because H is invariant and both hold on it, V_i equals the holding value of Lemma 1 on H. Let $g_i(z) = -(z - h_i)^2 + g V_i(z)$ be player i's one-round objective as a function of the next position. On H it is G_i of Lemma 2. At any z the policy moves the dog to $N(z)$, so $V_i(z) = g_i(N(z))$, and therefore

g_i(z) = -(z - h_i)^2 + g g_i(N(z)) for every z. (*)

Claim: g_red is nondecreasing on [0, T_red] and nonincreasing on [T_red, 1] (each part when nonempty).

Above the target. Let M be the set of bounded functions phi on [T_red, 1] that equal G_red on [T_red, X_red] and are nonincreasing. Define (A phi)(z) = G_red(z) on [T_red, X_red] and (A phi)(z) = -(z - h_red)^2 + g phi(N(z)) on R; this is well defined because N maps R into [T_red, 1). A is a contraction with modulus g in the sup norm, and by (*) the restriction of g_red to [T_red, 1] is its fixed point. A maps M into M:

- on R, -(z - h_red)^2 decreases, because z > X_red >= T_red >= h_red (T_red - h_red = (k - 1)(h_red - d*) >= 0), and phi(N(z)) does not increase, because N does not decrease and phi does not increase;
- across the boundary, for z1 in [T_red, X_red] and z2 in R, (A phi)(z2) <= -(z2 - h_red)^2 + g phi(T_red) < -(X_red - h_red)^2 + g G_red(T_red) = G_red(X_red) <= G_red(z1), using N(z2) >= T_red, phi(T_red) = G_red(T_red), the identity G_red(X_red) = -(X_red - h_red)^2 + g G_red(z_h(X_red)) with z_h(X_red) = T_red (this is (*) on H), and that G_red decreases after T_red.

M is closed under uniform limits and contains, for example, G_red on [T_red, X_red] extended by the constant G_red(X_red), so iterating A from it converges to the fixed point, which therefore lies in M (Banach's fixed point theorem).

Below the target. The same argument on [0, T_red] with the set of functions that equal G_red on [X_blue, T_red] and are nondecreasing: on B, -(z - h_red)^2 increases because z < X_blue <= d* <= h_red, and N maps B into (z, T_blue], which lies in [0, T_red]; across the boundary, (A phi)(z1) <= -(z1 - h_red)^2 + g G_red(T_blue) < -(X_blue - h_red)^2 + g G_red(T_blue) = G_red(X_blue) <= G_red(z2) for z1 in B and z2 in [X_blue, T_red], using z_h(X_blue) = T_blue, T_blue in [X_blue, T_red] and that G_red increases up to T_red. This proves the claim; the claim for blue is the mirror image, with its peak at T_blue.

Best replies. A function that does not decrease up to T and does not increase after it is maximized over any interval [L, U] at clip(T, L, U). Given blue's action at x, red reaches the next positions [U - (1 - w0) w, U], where U is the position with a_red = 1. On H and R, blue is at 0, so U = z_h(x) and red's best reply is clip(T_red, U - (1 - w0) w, U), which is exactly the policy (it holds on H, where z_h(x) <= T_red, and moves to max(T_red, z_h(x) - (1 - w0) w) on R). On B, blue moves the position up to N(x) <= T_blue <= T_red with red at 1, so U = N(x) <= T_red and the best reply is red's edge, as in the policy. Blue's case is the mirror image. By the one-shot deviation principle the policy is a feedback equilibrium. ■

Many houses

With N players, weights w_i summing to 1 and the dog at sum_i w_i a_i, Lemmas 1 and 2 hold unchanged for each player (the other players' held actions only enter through d*, and player i reaches an interval of length (1 - w0) w_i). So Proposition 1 holds verbatim: holding the one-shot actions is a feedback equilibrium iff every player at an edge has slack at least rho(w0, g) and, when w0 > 0, no player (the pivot) is strictly inside the box. In the one-shot equilibrium, which is the generalized median of Renault and Trannoy (2005), at most one player is inside. The study `many_houses` measures how often there is none, and how the largest admissible inertia scales with N.

Continuous-time limit

With time step dt, w0 = 1 - dt and g = exp(-r dt), 1 - g w0 = (r + 1) dt + O(dt^2) and 1 - g w0^2 = (r + 2) dt + O(dt^2), so k -> (r + 2) / (r + 1) and rho -> (r + 1) / (r + 2). In Proposition 2, (T_i - z_h(x)) / ((1 - w0) w_i) = (T_i - x + dt (x - d*)) / (dt w_i), which tends to +infinity where x < T_i and to -infinity where x > T_i, so the policy tends to the bang-bang rule "push the edge toward T_i", with the interior band of width O(dt). This describes the limit of the discrete-time equilibria; that it is an equilibrium of the continuous-time game is not proved here.

What the propositions do not cover

- **Interior one-shot actions.** If a player's one-shot action is strictly inside the box, its hold region is the single point d*, and off it that player pushes at an edge; the dog then returns faster than under holding, the continuation in Lemma 2 is wrong, and the other player's best target moves. In one random game (houses 0.754 and 0.538, w = 0.347, w0 = 0.767, g = 0.335) the dynamic program's red aims at about 0.785 where the policy of Proposition 2 aims at 0.771. The policy fails in 53 of 123 such random games; the equilibrium there is open.
- **Uniqueness.** The propositions give one feedback equilibrium; they do not rule out others.
- **Shared coordinates.** When the houses coincide, both one-shot actions can be interior and the one-shot game has a continuum of equilibria.